Algebra 2 — Unit 2: Quadratic Functions

Practice Test — Answer Key

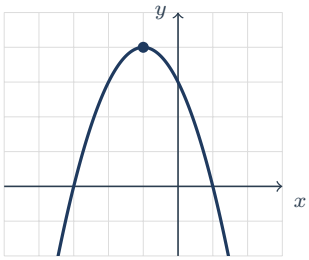
Instructions: Show all work. Give exact answers — simplest radical form, and $a + bi$ form for non-real numbers. No notes or calculator unless your teacher says so. **Total: 100 pts**

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Reading & Graphing Parabolas (20 pts)

1. (7 pts) The graph of a quadratic function f is shown. Read each feature from the graph.



Vertex: $(-1, 4)$

Axis of symmetry: $x = -1$

Maximum or minimum? **maximum**

Its value: 4

Zeros: -3 **and** 1

Increasing on: $x < -1$, $(-\infty, -1)$

Range: $y \le 4$, $(-\infty, 4]$

2. (3 pts) Which describes the graph of $y = -2(x-3)^2 + 5$? **Letter: C**

(A) opens up; vertex (3, 5)

(B) opens down; vertex (−3, 5)

→ **(C)** opens down; vertex (3, 5)

(D) opens up; vertex (−3, 5)

3. (5 pts) For $f(x) = 2x^2 + 8x + 3$, use $x = -\dfrac{b}{2a}$ to find the axis of symmetry and the vertex.

$$x = -\frac{b}{2a} = -\frac{8}{2(2)} = -2$$

$$f(-2) = 8 - 16 + 3 = -5$$

Axis: $x = -2$

Vertex: $(-2, -5)$

4. (5 pts) For $y = (x+4)(x-2)$, find the zeros, the axis of symmetry, and the vertex.

$$x + 4 = 0 \quad \text{or} \quad x - 2 = 0$$

$$x = \frac{-4 + 2}{2} = -1$$

$$y = (-1 + 4)(-1 - 2) = -9$$

Zeros: -4 **and** 2

Axis: $x = -1$

Vertex: $(-1, -9)$

Part B — Solving by Factoring (20 pts)

Factor completely: take out the GCF first. To solve, get the equation equal to zero, factor, and set each factor equal to zero.

5. (8 pts) Factor completely.

   (a) $3x^2 - 12x - 36$

   $$\mathbf{3x^2 - 12x - 36 = 3(x^2 - 4x - 12)}$$

   $3(x-6)(x+2)$

   (b) $x^2 - 81$ — $(x+9)(x-9)$

   (c) $x^2 - 10x + 25$ — $(x-5)^2$

   (d) $3x^2 + 4x - 4$ (by grouping)

   $$\mathbf{3x^2 + 4x - 4 = 3x^2 + 6x - 2x - 4}$$
   $$\mathbf{= 3x(x+2) - 2(x+2)}$$

   $(3x-2)(x+2)$

6. (4 pts) Solve $x^2 + 3x - 28 = 0$.

   $$\mathbf{(x+7)(x-4) = 0}$$
   $$\mathbf{x + 7 = 0 \quad \text{or} \quad x - 4 = 0}$$

   Solution: $x = -7$ **or** $x = 4$

7. (4 pts) Solve $2x^2 = 10x$.

   $$\mathbf{2x^2 - 10x = 0}$$
   $$\mathbf{2x(x-5) = 0}$$

   Solution: $x = 0$ **or** $x = 5$

8. (4 pts) Solve $(x+2)(x-3) = 6$.

   $$\mathbf{x^2 - x - 6 = 6}$$
   $$\mathbf{x^2 - x - 12 = 0}$$
   $$\mathbf{(x-4)(x+3) = 0}$$

   Solution: $x = 4$ **or** $x = -3$

Part C — Square Roots & Complex Numbers (20 pts)

Get the square alone before you take the root, and keep the $\pm$. Write non-real numbers in $a + bi$ form.

9. (3 pts) Solve $3x^2 + 5 = 65$.

   $$\mathbf{3x^2 = 60}$$
   $$\mathbf{x^2 = 20}$$
   $$\mathbf{x = \pm\sqrt{20}}$$

   Solution: $x = \pm 2\sqrt{5}$

10. (4 pts) Solve $(x-3)^2 = 12$.

    $$\mathbf{x - 3 = \pm\sqrt{12}}$$
    $$\mathbf{x - 3 = \pm 2\sqrt{3}}$$

    Solution: $x = 3 \pm 2\sqrt{3}$

11. (4 pts) (a) Simplify $\sqrt{-72}$: $6i\sqrt{2}$ (b) Solve $x^2 + 49 = 0$: $x = \pm 7i$

12. (6 pts) Let $z_1 = 4 + 3i$ and $z_2 = 2 - 5i$. Write each answer in $a + bi$ form.

(a) $z_1 + z_2 =$ $6 - 2i$ (b) $z_1 - z_2 =$ $2 + 8i$

(c) $z_1 \cdot z_2$

$$(4 + 3i)(2 - 5i) = 8 - 20i + 6i - 15i^2$$
$$= 8 - 14i + 15$$

$z_1 \cdot z_2 =$ $23 - 14i$

13. (3 pts) $\sqrt{-4} \cdot \sqrt{-25} =$ -10 (A) 10 → **(B) −10** (C) $10i$ (D) $-10i$

Part D — Completing the Square, Formula & Discriminant (20 pts)

14. (4 pts) Solve by completing the square: $x^2 + 8x - 5 = 0$

$$x^2 + 8x = 5$$
$$x^2 + 8x + 16 = 5 + 16$$
$$(x + 4)^2 = 21$$
$$x + 4 = \pm\sqrt{21}$$

Solution: $x = -4 \pm \sqrt{21}$

For 15 and 16, use the quadratic formula. Write the equation in standard form first.

15. (5 pts) Solve $2x^2 + 1 = 6x$.

$$2x^2 - 6x + 1 = 0$$
$$x = \frac{6 \pm \sqrt{36 - 8}}{4}$$
$$= \frac{6 \pm \sqrt{28}}{4}$$

Solution: $x = (3 \pm \sqrt{7})/2$

16. (5 pts) Solve $x^2 + 6x + 13 = 0$.

$$x = \frac{-6 \pm \sqrt{6^2 - 4(1)(13)}}{2(1)}$$
$$= \frac{-6 \pm \sqrt{-16}}{2} = \frac{-6 \pm 4i}{2}$$

Solution: $x = -3 \pm 2i$

17. (6 pts) For each equation, find the discriminant $b^2 - 4ac$. Then give the number and type of solutions (*two real*, *one repeated real*, or *two complex*) and the number of x-intercepts of its graph.

Equation	Discriminant	Solutions	x-intercepts
$x^2 - 6x + 9 = 0$	0	one repeated real	1
$2x^2 + 3x + 4 = 0$	-23	two complex	0
$x^2 + 5x - 2 = 0$	33	two real	2

Part E — Systems & Modeling (20 pts)

Give each solution of a system as an ordered pair. Answer each model question with units.

18. (7 pts) Solve the system by substitution.

$$y = x^2 - 4x + 1 \qquad y = x - 3$$

$$x^2 - 4x + 1 = x - 3$$
$$x^2 - 5x + 4 = 0$$
$$(x-1)(x-4) = 0$$
$$x = 1: \quad y = 1 - 3 = -2$$
$$x = 4: \quad y = 4 - 3 = 1$$

Solutions: [(1, −2) and (4, 1)]

19. (3 pts) How many points do the graphs of $y = x^2 + 1$ and $y = 2x$ have in common?

(A) none

→ **(B)** exactly one

(C) exactly two

(D) infinitely many

Letter: [B]

20. (10 pts) At the egg drop, an egg is launched straight up from the roof of the science building. Its height, in feet, t seconds after launch is

$$h(t) = -16t^2 + 48t + 64.$$

(a) (2 pts) How high is the roof?

[64 ft]

(b) (4 pts) When does the egg reach its greatest height, and what is that height?

$$t = -\frac{48}{2(-16)} = 1.5$$
$$h(1.5) = -36 + 72 + 64 = 100$$

When: [1.5 s] Height: [100 ft]

(c) (4 pts) When does the egg hit the ground? Which solution do you reject, and why?

$$-16t^2 + 48t + 64 = 0$$
$$t^2 - 3t - 4 = 0$$
$$(t-4)(t+1) = 0$$

Lands at: [$t = 4$ s]

Rejected: [$t = -1$: negative time]